

Jiaqi Li

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EDUCATION

Southeast University (Nanjing, China)

B.S. in Computer Science and Technology

GPA: 90.89 / 100

Honors: 6th Global Campus AI Algorithm Elite Competition (3rd); President's Scholarship; Software Innovation Scholarship; Outstanding Student Award

Sep 2022 - June 2026

University of Michigan (Ann Arbor, USA)

M.S. in Computer Science and Engineering

Sep 2026 - current

PUBLICATIONS

- Gao, Y., Li, J.-Q., & Zhang, M.-L. (2025). A denoising framework with exclusivity constraints for partial multi-label learning. *Chinese Science: Information Sciences. (CCF-A)* <https://www.sciengine.com/SSI/doi/10.1360/SSI-2025-0525>
- Song, X., Yin, W., Li, J., Wei, C., Wang, Y., Pi, J., Chen, L., Gao, C., Ma, W., & Wu, J. (2025). Recognition and diagnosis of acute ischemic stroke with large vessel occlusion based on ChatGLM. *Chinese Journal of Stroke*, 20(1), 70–77. <https://doi.org/10.3969/j.issn.1673-5765.2025.01.009>

RESEARCH EXPERIENCES

Multimodal Pretraining of Fundus Image Foundation Models with Paired Image-Text Data Sep 2023 – Oct 2024

Advisor: Prof. Yi Zhou

- Constructed a high-quality, precisely aligned image–text dataset (MM-Retinal) with about 30,000 samples, based on five illustrated ophthalmology textbooks, and covering a wide range of fundus pathologies.
- Designed a FLAIR-based multimodal pretraining approach, using ResNet50 and BioClinicalBERT to encode images and text during image–text pair generation, optimizing cross-modal alignment.
- Inspired by Tip-Adapter, proposed a lightweight expert knowledge fusion strategy that enhances text prompts by identifying image similarity and retrieving prior knowledge, and improves reasoning robustness and diagnostic accuracy on public datasets lacking embedded domain knowledge.
- Adopted a mixed training strategy to balance the sample distribution between MM-Retinal and open datasets, improving model generalization, expert knowledge adaptation, and multimodal diagnostic accuracy and stability.

3D Reconstruction and Segmentation Based on Cryo-Electron Tomography Images

Jul 2024 – Aug 2024

Advisor: Prof. Zhaoqian Wang, Westlake University

- Developed deep learning methods to automate the analysis of Cryo-ET images for molecular-level studies of viral structures in structural biology.
- Deployed state-of-the-art segmentation models such as SAM, to enhance the efficiency and accuracy of biological feature segmentation.
- Optimized the training process and enhanced the model architecture to provide structural biologists with an effective tool to study viral replication.

ChatGLM-Based Identification and Diagnosis of Large Vessel Occlusion in Acute Ischemic Stroke

Oct 2024 – Jun 2025

Advisor: Professor Ma Weizhi, Tsinghua University

- Built a Large Vessel Occlusion (LVO) identification task using 935 clinical cases of acute ischemic stroke (including 230 LVO cases) from Tsinghua Changgeng Hospital (2016–2024), with labels grouped by CTA/MRA/DSA imaging results.
- Developed five reasoning strategies, including zero-shot, zero-shot with chain-of-thought (CoT), pretrained language model with CoT, few-shot, and few-shot with CoT, based on ChatGLM, integrating prompt engineering, instruction tuning, and retrieval-augmented generation (RAG) to generate reasoning paths and diagnostic outcomes.
- Achieved significant improvements in diagnostic performance, with accuracy increasing from 36.1% to 75.1%, along with notable gains in sensitivity and specificity.

Denoising Framework with Exclusivity Constraints for Partial Multi-Label Learning

Oct 2024 – Mar 2026

Advisor: Dean Zhang Minling, Southeast University,

- Redesigned the loss function and incorporated label propagation methods into a weakly supervised learning framework to address partial multi-label classification problems.
- Enabled a multi-label denoising framework using distributed attention and suppressing noisy signals, outperforming state of the art models across 11 benchmarks.
- Method was tested and evaluated on 11 benchmark datasets, and the results show that it outperforms five state-of-the-art baseline models, demonstrating its effectiveness in reducing label noise and improving generalization ability.

PROJECTS

Academicbrain Web Development

Feb 2025 – June 2025

- Developed a web platform targeted at researchers and students in computer science, designed to provide efficient, personalized paper recommendation, summarization, and management.
- Built the system architecture using Spring Boot, MySQL for the backend, and Vue.js/Tailwind CSS for the frontend, deployed on Tencent Cloud.
- Core features included a personalized recommendation system based on collaborative filtering and content-based algorithms; data collection via web crawlers and public APIs (arXiv, OpenReview, Google Scholar); paper summaries with AI-generated insights; interactive comment sections; citation visualization; and user profiles with folder management, reading statistics, and behavioral analytics.
- Conducted research and contributed to the design of the recommendation algorithms. Took primary responsibility for frontend structure and interaction design, implementing and integrating the homepage, paper display, and user profile modules to ensure a unified interface and seamless user experience.
- Authored requirement analysis and operational documentation to ensure standardized development workflows and maintainability.

Fine-tuning Large Language Models on TCM Datasets (Remote Project, CMU)

Feb 2024 – Oct 2024

- Fine-tuned Llama 3, Phi-3.5, and Mistral models on a TCM dataset to enhance their understanding of TCM terminology, diagnostic reasoning, and therapeutic concepts.
- Accelerated the training process using Unsloth, applied LoRA for efficient parameter optimization, and formatted the dataset to align with the Hugging Face framework.

RISC V System Co Design Project

Sep 2025 - Jan 2026

- Contributed to a five stage RV32IMC CPU and SoC implementation, including compressed instruction support, static branch prediction, data forwarding, load use hazard handling, and basic exception and interrupt control.
- Developed the complete software toolchain, including a Mini C compiler with front end, back end, and register allocation, along with the runtime library and BIOS firmware for initialization, interrupt handling, and program loading.
- Designed and implemented an integrated development environment that supported code editing, compilation, debugging, and deployment for a full bare metal workflow.
- Delivered stable application execution on FPGA hardware with reliable system startup, interrupt behavior, and user mode operation.

SKILLS

Programming skills and Tools : Java, C#, Python, C++, SQL, .NET Framework, Spring Boot, [Vue.js](#), Tailwind CSS, Hugging Face, Cloud computing platforms like AWS, Microsoft Azure, GCP and Tencent Cloud.

Language: Chinese(Native), English